Refugee Movement Around the World

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Overview

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Introduction

Supporting smarter refugee aid planning and migration policy.

- Goal
  - Uncover key refugee flow patterns using graph analytics
- Approach
  - Model refugee flows with Neo4j
  - Use graph algorithms to find influential countries and transit routes



MongoDB

Storing Refugee Profiles and Case Histories

- Flexible, document-based NoSQL database
- Ideal for diverse, evolving refugee records
- Handles inconsistent data across regions
- Supports real-time updates and schema changes
- Simpler than relational DBs with no complex joins

Redis

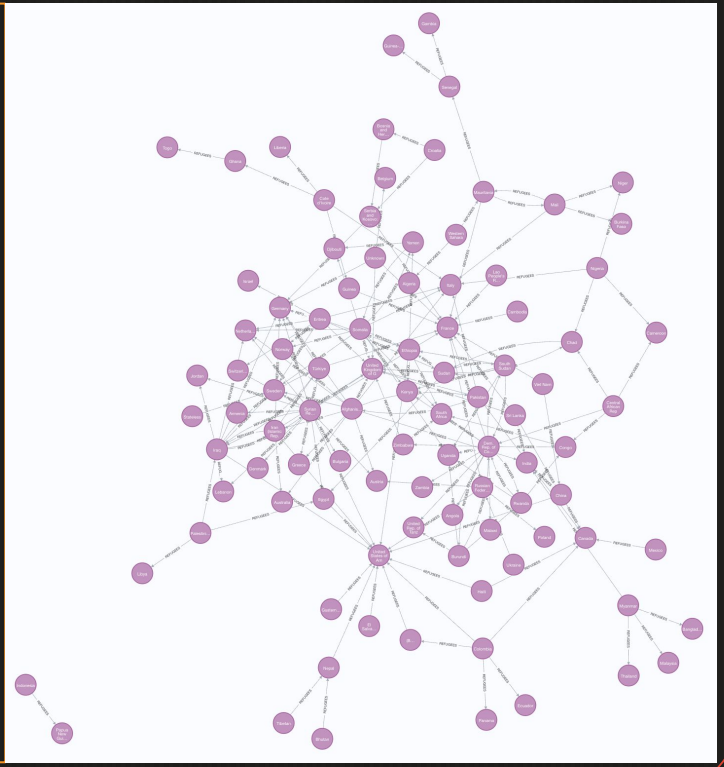
Real-Time Tracking and Monitoring Dashboards

- In-memory key-value store optimized for speed
- Real-time border crossing alerts and dashboards
- Tracks live supply counts (beds, meals, supplies)
- Instant updates for time-sensitive scenarios
- Outperforms relational DBs in real-time use

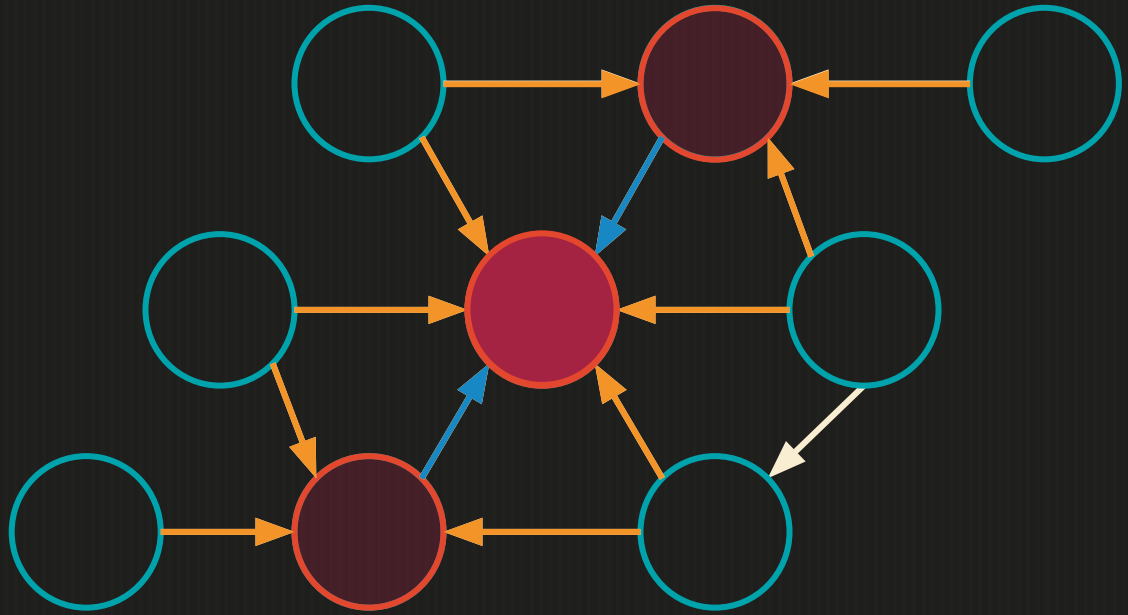
Neo4j

Analyzing Refugee Movement Around the World

- Countries as nodes, refugee flows as directed, weighted edges
- Captures complex migration patterns from origin to asylum
- Enables analysis with graph algorithms
- More effective than relational databases for network analysis



What is Page Rank?



Depth & Damping factor

Page Rank in Our Context

How to interpret a ‘high-influence’ country?



Primary Refugee Recipients

(TWO DIFFERENT GRAPHS)



Primary Refugee Producers

OVERALL PAGE RANK

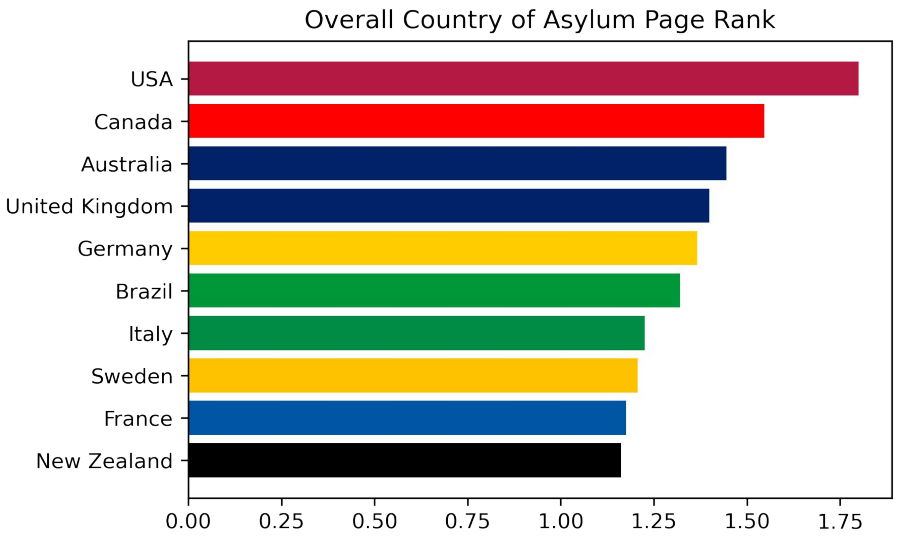
Two graphs:

country of origin
↓
country of asylum

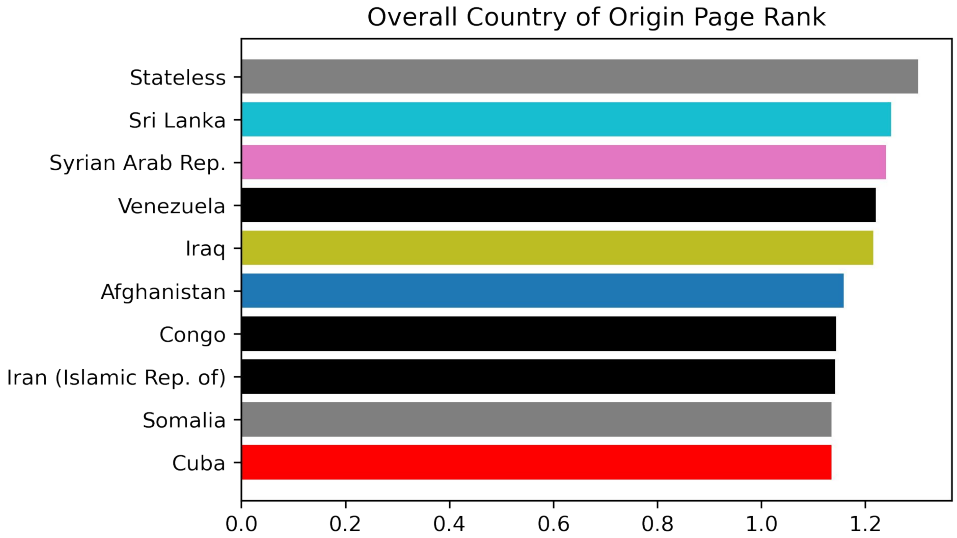
country of origin
↑
country of asylum

OVERALL - PAGE RANK

COUNTRIES OF ASYLUM

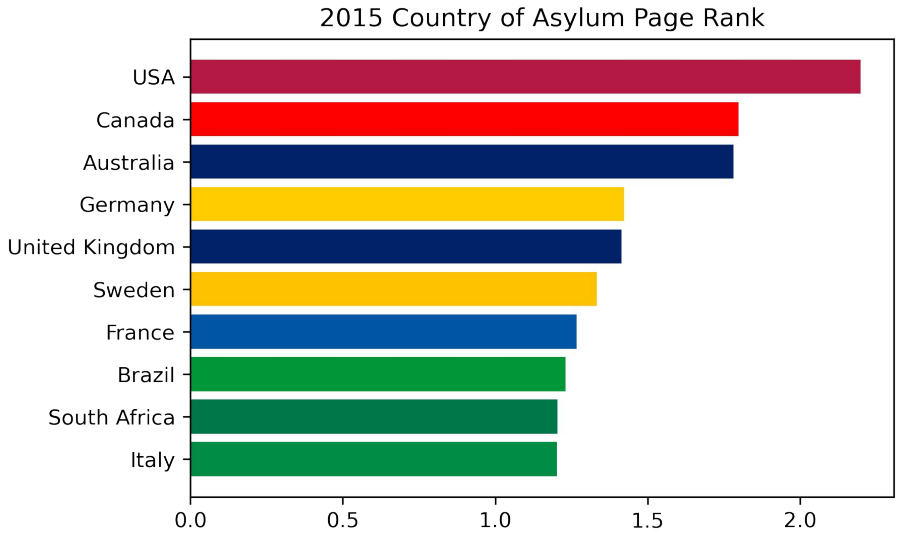


COUNTRIES OF ORIGIN

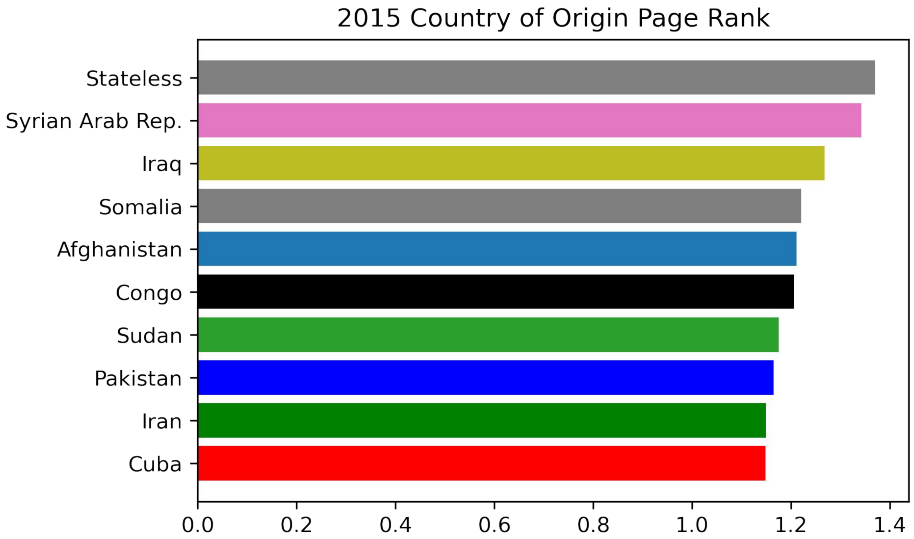


2015 - PAGE RANK

COUNTRIES OF ASYLUM

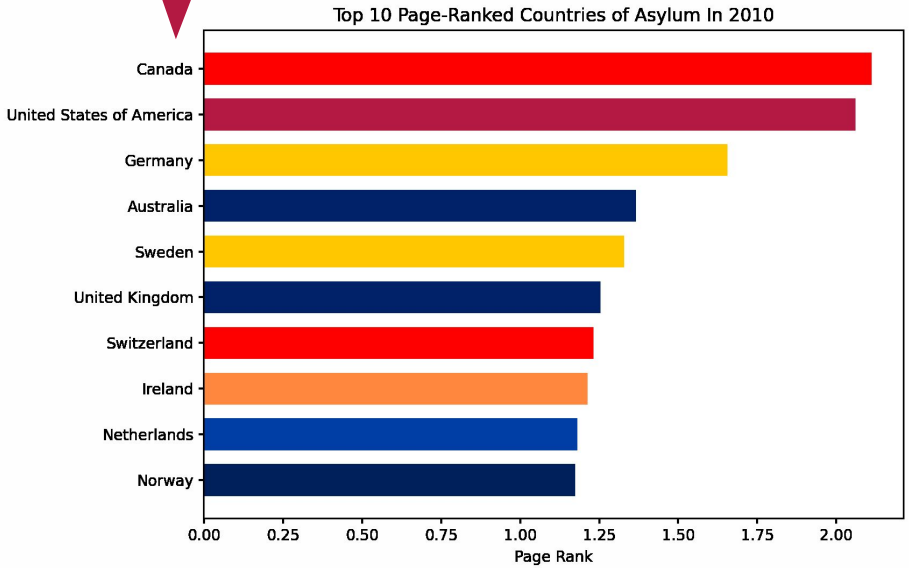


COUNTRIES OF ORIGIN

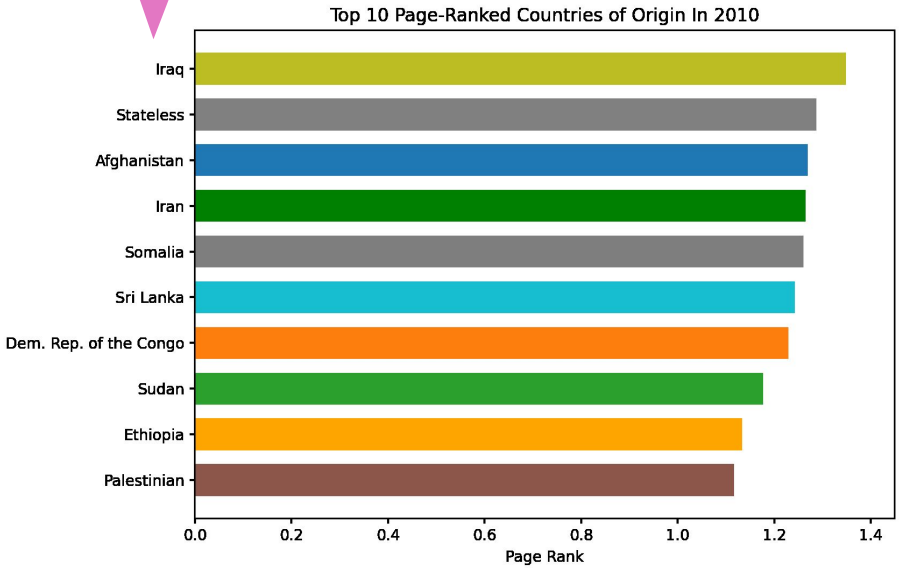


OVER TIME - PAGE RANK

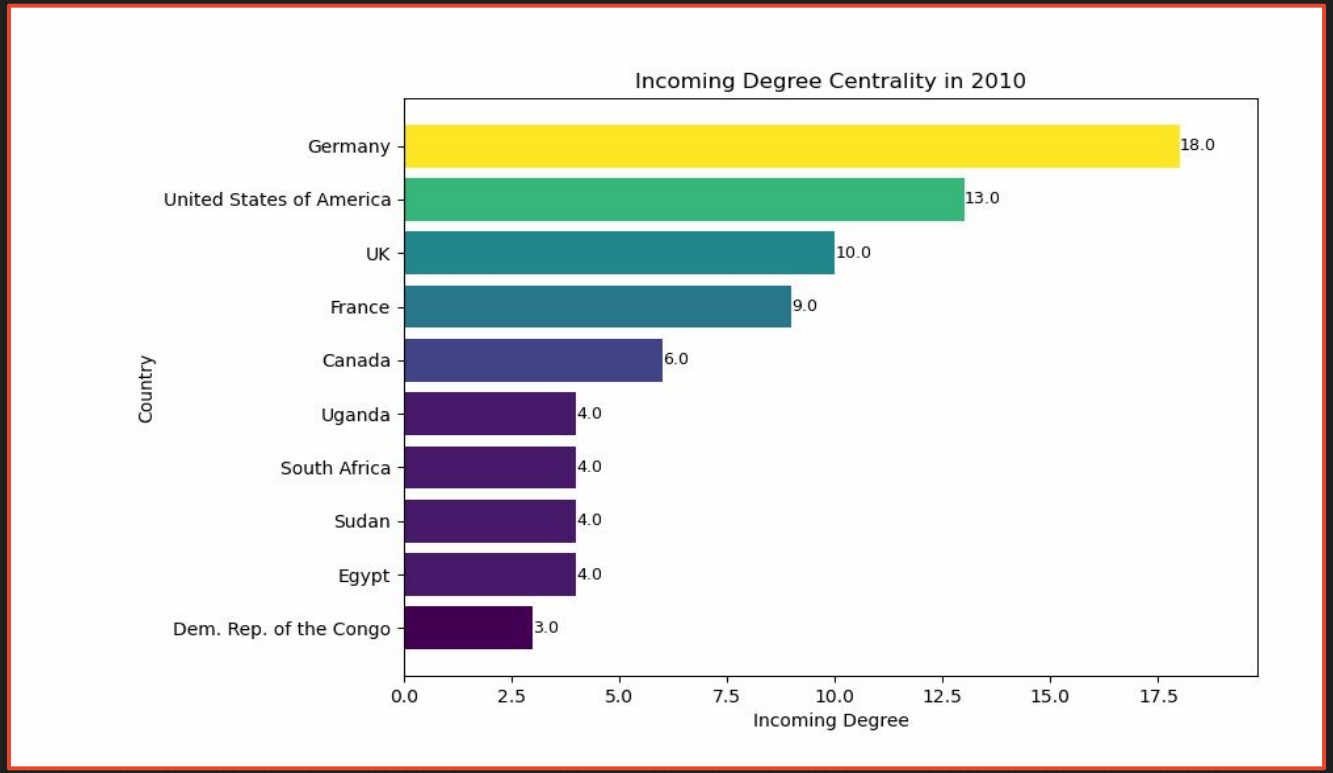
COUNTRIES OF ASYLUM



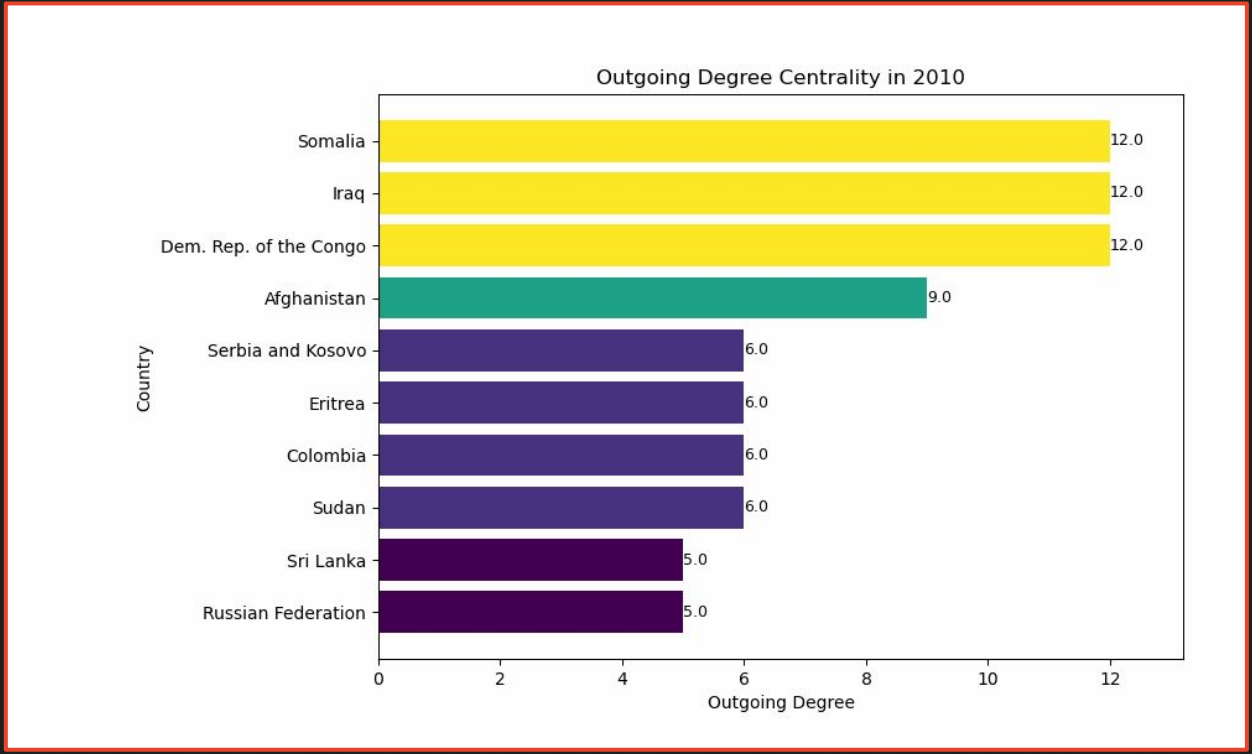
COUNTRIES OF ORIGIN



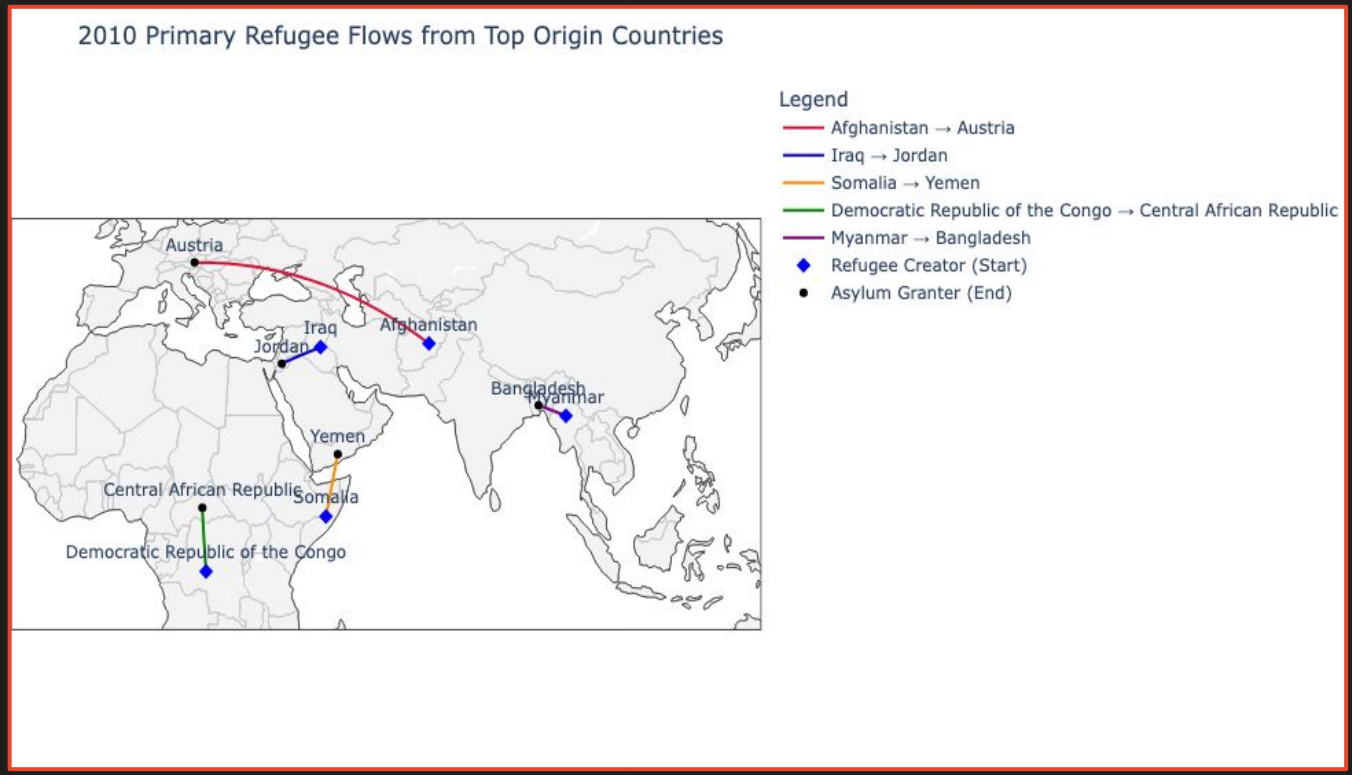
Degree Centrality - Incoming (2010/2015/2020)



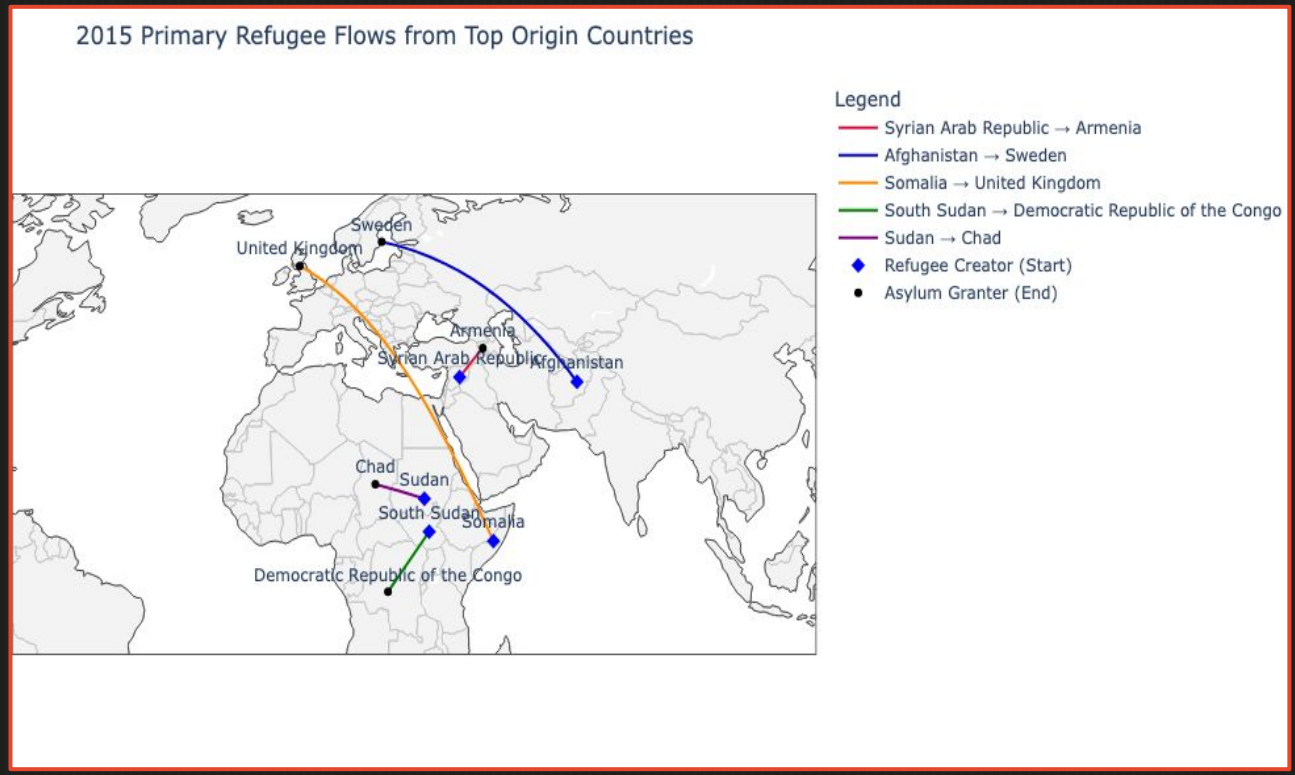
Degree Centrality - Outgoing (2010/2015/2020)



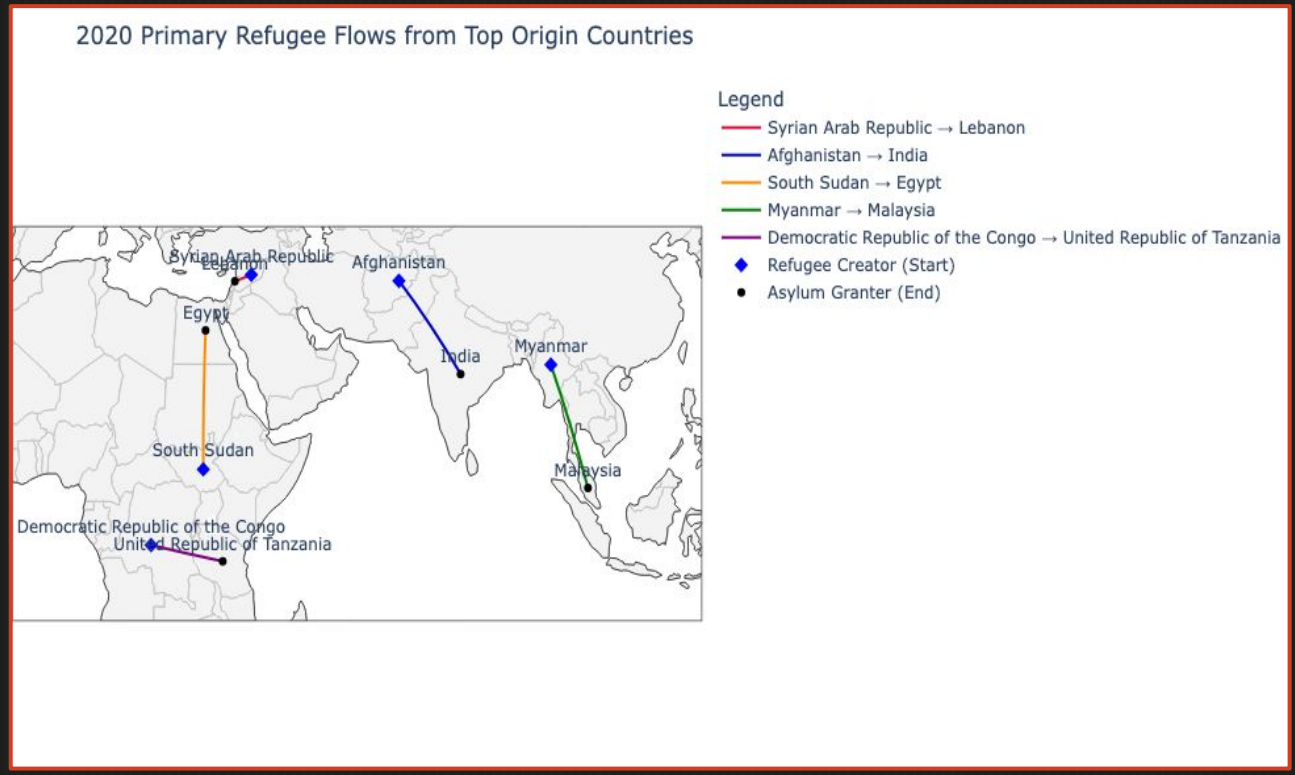
Min. Spanning Tree 2010



Min. Spanning Tree 2015

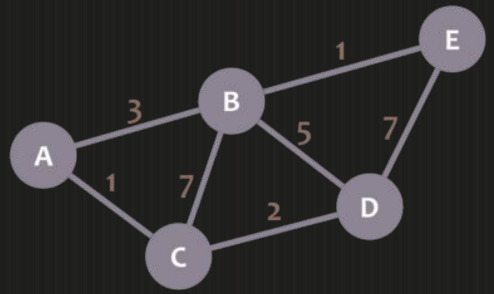


Min. Spanning Tree 2020

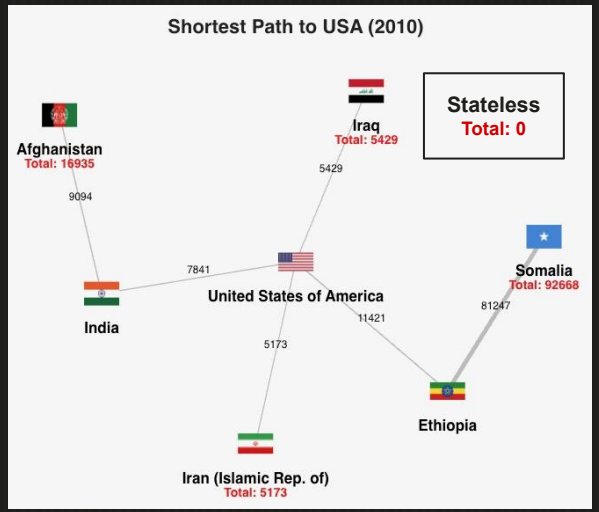


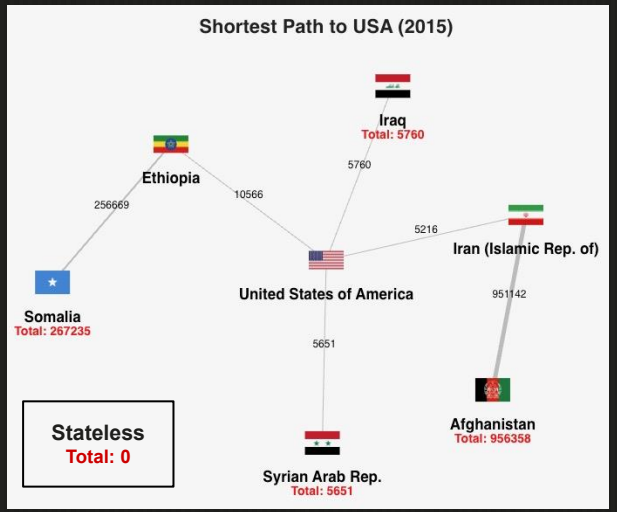
Shortest Path

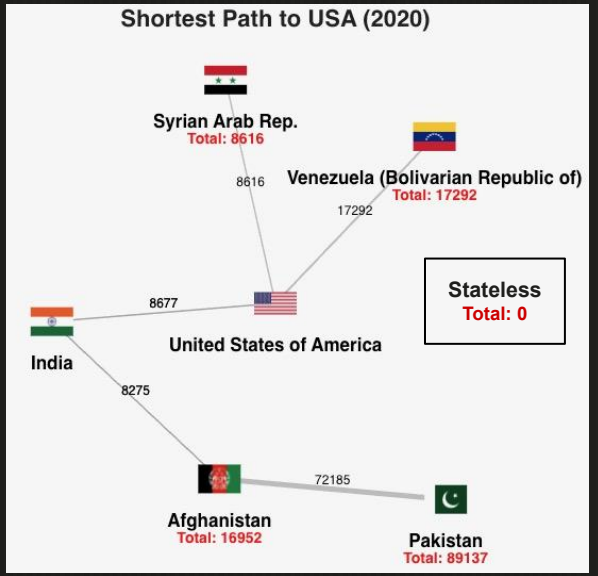
- Determine how *direct* or *indirect* refugee movements are, particularly towards the U.S.A.
- Dijkstra's Algorithm → Find shortest path from each COO to U.S.A. by *minimizing cost* and number of *intermediate countries*
- *Page Rank* → Top 5 Countries of Origin for 2010, 2015, and 2020
- Help operational aid planning, policy targeting, and refugee support



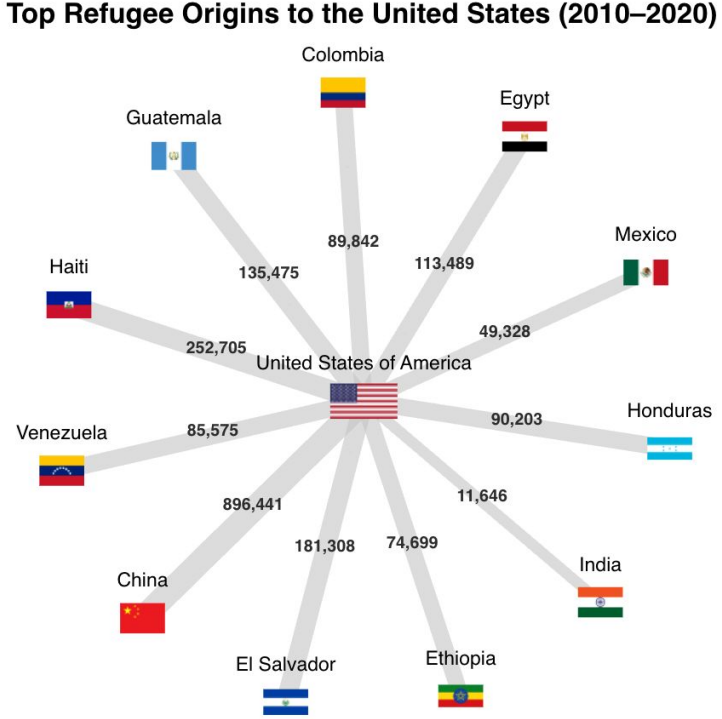
Shortest Path







Conclusion


Top Refugee Origins to the United States (2010–2020)
Colombia
Egypt
Guatemala
Mexico
Haiti
89,842
135,475
113,489
49,328
252,705
United States of America
Honduras
90,203
Venezuela
85,575
896,441
11,646
181,308
74,699
India
China
El Salvador
Ethiopia


THANK YOU